

Homework 3.4

Problem 11

If f' is continuous and positive on $[a, b]$, and if $f(a)f(b) < 0$, then f has exactly one zero in (a, b) .
Prove this, and show that with a suitable parameter λ , the zero can be obtained by applying the method of functional iteration, with $F(x) = x + \lambda f(x)$.

First to show that there exists only one zero.

$$\begin{aligned} f'(x) &\in C[a, b], f'(x) > 0, \forall x \in [a, b] \\ \Rightarrow \int_a^x f'(\chi) d\chi &= f(x) - f(a) \text{ is strictly increasing on } [a, b] \\ f(a)f(b) < 0 &\Rightarrow f(a) \in \mathbb{R}^-, f(b) \in \mathbb{R}^+ \end{aligned}$$

Using the fact that the antiderivative of a continuous function is continuous and the Intermediate Value Theorem, we get that $\exists c \in (a, b)$ such that $f(c) = 0$ and since the function is strictly increasing, we get that this value is unique.

For $F(x)$ to have a fixed point, we need that $F(x)$ be contracting:

$$|F(x) - F(y)| = |x - \lambda f(x) - y + \lambda f(y)| = |x - y - \lambda(f(x) - f(y))| \leq \tilde{\lambda}|x - y|$$

Using mean value theorem:

$$f'(c) = \frac{f(b) - f(a)}{b - a} > 0 \Rightarrow f(x) - f(y) = f'(c)(x - y)$$

Making $\frac{1}{\lambda} = M = \max\{f'(x) | x \in [a, b]\}$, we would get that:

$$|x - y - \lambda(f(x) - f(y))| \leq \left| x - y - \frac{f'(c)}{M}(x - y) \right| = \left| 1 - \frac{f'(c)}{M} \right| |x - y|$$

Noting the following:

$$\left| 1 - \frac{f'(c)}{M} \right| \leq 1$$

We get what we desired.

$$\Rightarrow |F(x) - F(y)| \leq \tilde{\lambda}|x - y|$$

Problem 20

Show that the following functions are contractive on the given domain, yet they have no fixed point on these domains. Why does this not contradict the Contractive Mapping Theorem?

a) $F(x) = 3 - x^2, \left[-\frac{1}{4}, \frac{1}{4}\right]$

$$|F(x) - F(y)| = |3 - x^2 - 3 + y^2| = |y^2 - x^2| = |x + y||x - y|$$

$$\max_{x,y \in \left[-\frac{1}{4}, \frac{1}{4}\right]} \{|x + y|\} = \frac{1}{4} \Rightarrow |x + y||x - y| \leq \frac{1}{4}|x - y|$$

$$\therefore |F(x) - F(y)| \leq \frac{1}{4}|x - y|, F: \left[-\frac{1}{4}, \frac{1}{4}\right] \rightarrow \left[3 - \frac{1}{8}, 3\right]$$

b) $F(x) = -\frac{x}{2}, [-2, -1] \cup [1, 2]$

$$|F(x) - F(y)| = \left| -\frac{x}{2} - \left(-\frac{y}{2}\right) \right| = \frac{1}{2}|y - x| = \frac{1}{2}|x - y|$$

$$F: [2, -1] \cup [1, 2] \rightarrow \left[-1, -\frac{1}{2}\right] \cup \left[\frac{1}{2}, 1\right]$$

Since neither is an into mapping, we do not satisfy the Theorem.

Problem 26

If the method of functional iteration is used on $f(x) = \frac{1}{2}(1+x)^{-1}$ starting at $x_0 = 7$, will the resulting sequence converge? If so, what is the limit? Establish your answer rigorously.

$$|f(x) - f(y)| = \frac{1}{2} \left| \frac{1}{1+x} - \frac{1}{1+y} \right| = \frac{1}{2} \left| \frac{1+y-1-x}{(1+x)(1+y)} \right| = \frac{1}{2} \left| \frac{y-x}{1+x+y+xy} \right|$$

Since we can see that the output is going to be positive if the x values are positive and we started with a positive value, then we have that $x, y \geq 0$ which means that

$$\frac{1}{2} \left| \frac{y-x}{1+x+y+xy} \right| \leq \frac{1}{2}|x-y|$$

Thus, we know that this function is contracting.

Since we have that all our inputs in this iteration will be greater than 0, we have that $f: \mathbb{R}^+ \rightarrow \left(0, \frac{1}{2}\right)$

Thus we have that this is an into mapping, therefore our theorem guarantees a solution for any positive input.

since we have that $\lim_{n \rightarrow \infty} x_{n+1} = \lim_{n \rightarrow \infty} \frac{1}{2}(1+x_n)^{-1} = s$, then we can do the following:

$$s = \frac{1}{2} \left(\frac{1}{1+s} \right) \Rightarrow 2s = \frac{1}{1+s} \Rightarrow 2s(1+s) = 1 \Rightarrow 2s^2 + 2s - 1 = 0 \Rightarrow s^2 + s - \frac{1}{2} = 0$$

$$s = \frac{-1 \pm \sqrt{1 - 4(1)\left(-\frac{1}{2}\right)}}{2(1)} = \frac{-1 \pm \sqrt{3}}{2}$$

But since we said that our solution is positive:

$$s = \frac{-1 + \sqrt{3}}{2} \approx .366025$$

Homework 6.1

Problem 1

Find the polynomial of least degree that interpolates these sets of data.

x	3	7	1	2
y	10	146	2	1

$$p(x) = \sum_{i=0}^n p(x_i) \ell_i(x), \ell_i(x) = \prod_{\substack{j=0 \\ i \neq j}}^n \frac{x - x_j}{x_i - x_j}$$

$$p(x) = 10 * \frac{(x-7)(x-1)(x-2)}{(3-7)(3-1)(3-2)} + 146 * \frac{(x-3)(x-1)(x-2)}{(7-3)(7-1)(7-2)} + 2 * \frac{(x-3)(x-7)(x-2)}{(1-3)(1-7)(1-2)} + \frac{(x-3)(x-7)(x-1)}{(2-3)(2-7)(2-1)}$$

$$\therefore p(x) = -\frac{10}{8}(x-7)(x-1)(x-2) + \frac{146}{120}(x-3)(x-1)(x-2) - \frac{1}{6}(x-3)(x-7)(x-2) + \frac{1}{5}(x-3)(x-7)(x-1)$$

$$p(3) = -\frac{10}{8} * (-4 * 2 * 1) = 10$$

$$p(7) = \frac{146}{120} * (4 * 6 * 5) = 146$$

$$p(1) = -\frac{1}{6} * (-2 * -6 * -1) = 2$$

$$p(2) = \frac{1}{5} * (-1 * -5 * 1) = 1$$

Problem 9

Prove that if g interpolates the function f at $x_0, x_1, \dots, x_{n-1}$ and if h interpolates f at $x_1, x_2, \dots, x_n$, then the function

$$K(x) = g(x) + \frac{x_0 - x}{x_n - x_0} [g(x) - h(x)]$$

Interpolates f at $x_0, x_1, \dots, x_n$. Notice that h and g need not be polynomials.

We know that $f(x_i) = g(x_i), i = 0, 1, \dots, n$ and $f(x_j) = h(x_j), j = 1, 2, \dots, n$, thus we can deduce that $g(x_k) = f(x_k) = h(x_k), k = 1, 2, \dots, n-1$, thus:

$$K(x_k) = g(x_k) + \frac{x_0 - x_k}{x_n - x_0} [g(x_k) - h(x_k)] = f(x_k) + \frac{x_0 - x_k}{x_n - x_0} [f(x_k) - f(x_k)] = f(x_k)$$

Thus, we have that the $K(x_k) = f(x_k), k = 1, 2, \dots, n-1$. Now to check the endpoints.

$$K(x_0) = g(x_0) + \frac{x_0 - x_0}{x_n - x_0} [g(x_0) - h(x_0)] = g(x_0) = f(x_0)$$

$$K(x_n) = g(x_n) + \frac{x_0 - x_n}{x_n - x_0} [g(x_n) - h(x_n)] = g(x_n) - [g(x_n) - f(x_n)] = f(x_n)$$

$$\therefore K(x_l) = f(x_l), l = 0, 1, 2, \dots, n$$

Problem 13

Prove that if we take any set of 23 nodes on the interval $[-1, 1]$ and interpolate then following function $f(x) = \cosh(x)$ with a polynomial p of degree 22, then the relative error $|p(x) - f(x)| * [|f(x)|]^{-1}$ is no greater than $5 * 10^{-16}$, on $[-1, 1]$.

Using the Theorem on Polynomial Interpolating error for $n = 22$:

$$|f(x) - p(x)| = \left| \frac{1}{(n+1)!} f^{n+1}(\xi_x) * \prod_{i=0}^n (x - x_i) \right|$$

Also noting that since $(x - x_i) \leq 2$

$$|f(x) - p(x)| = \left| \frac{1}{23!} f^{23}(\xi_x) \right| \left| \prod_{i=0}^n (x - x_i) \right| \leq |f^{23}(\xi_x)| * \frac{2^{23}}{23!}$$

$$f(x) = \cosh(x) \Rightarrow f'(x) = \sinh(x) \Rightarrow f''(x) = \cosh(x) \Rightarrow \dots \Rightarrow f^{(23)}(x) = \sinh(x)$$

$$\sinh(-1) \leq \sinh(x) \leq \sinh(1) \Rightarrow |\sinh(x)| \leq \sinh(1)$$

$$\Rightarrow |f(x) - p(x)| \leq \frac{2^{23} * \sinh(1)}{23!}$$

$$1 \leq \cosh(x) \leq \cosh(1), x \in [-1, 1]$$

$$\therefore \frac{|f(x) - p(x)|}{|f(x)|} \leq \frac{2^{23} * \sinh(1)}{23!} \approx 3.8 * 10^{-16} < 5 * 10^{-16}$$

Problem 22

Find the Lagrange and Newton forms of the interpolating polynomial for these data:

x	-2	0	1
$f(x)$	0	1	-1

Write the polynomials in the form $a + bx + cx^2$ to verify that they are identical as functions.

Lagrange:

$$p(x) = 0 * \left(\frac{(x+2)(x-1)}{(0+2)(0-1)} \right) + 1 * \left(\frac{(x+2)(x-1)}{(1+2)(1-0)} \right) - 1 * \left(\frac{(x+2)(x-0)}{(1+2)(1-0)} \right)$$

$$p(x) = -\frac{1}{2}(x+2)(x-1) - \frac{1}{3}(x+2)(x) = -\frac{1}{2}(x^2 + x - 2) - \frac{1}{3}(x^2 + 2x)$$

$$p(x) = -\frac{5}{6}x^2 - \frac{7}{6}x + 1$$

Newton:

$$p_0(x) = 0$$

$$p_1(x) = 0 + c_1(x + 2) \Rightarrow p_1(0) = 1 = c_1(0 + 2) \Rightarrow c_1 = \frac{1}{2}$$

$$p_2(x) = \frac{1}{2}(x + 2) + c_2(x + 2)(x) \Rightarrow p_2(1) = -1 = \frac{1}{2}(1 + 2) + c_2(1 + 2)(1)$$

$$\Rightarrow -\frac{5}{2} = c_2 * 3 \Rightarrow c_2 = -5/6$$

$$p_2(x) = \frac{1}{2}(x + 2) - \frac{5}{6}(x^2 + 2x) = \frac{x}{2} + 1 - \frac{5}{6}x^2 - \frac{5}{3}x$$

$$\therefore p_2(x) = -\frac{5}{6}x^2 - \frac{7}{6}x + 1$$

$$p_2(-2) = -\frac{5}{6}(4) - \frac{7}{6}(-2) + 1 = -\frac{10}{3} + \frac{7}{3} + 1 = -1 + 1 = 0$$

$$p(0) = 1$$

$$p(1) = -\frac{5}{6} - \frac{7}{6} + 1 = -2 + 1 = -1$$

Problem 27

If we interpolate the function $f(x) = e^{x-1}$ with a polynomial p of degree 12 using 13 nodes in $[-1, 1]$, what is a good upper bound for $|f(x) - p(x)|$ on $[-1, 1]$.

$$|f(x) - p(x)| = \left| \frac{1}{(n+1)!} f^{(n+1)}(\xi_x) * \prod_{i=0}^n (x - x_i) \right|$$

$$(x - x_i) \leq 2 \Rightarrow \prod_{i=0}^n (x - x_i) \leq 2^{n+1}$$

$$f(x) = e^{x-1} \Rightarrow f'(x) = e^{x-1} \Rightarrow \dots \Rightarrow f^{(n+1)}(x) = e^{x-1}$$

$$\frac{1}{e^2} \leq |e^{x-1}| \leq 1, x \in [-1, 1]$$

$$\therefore |f(x) - p(x)| \leq \frac{2^{13}}{13!} \approx 1.3 * 10^{-6}$$

Homework 6.2

Problem 7

The divided difference $f[x_0, x_1]$ is analogous to a first derivative, as indicated in theorem 4. Does it have a property analogous to $(fg)' = f'g + g'f$?

$$f[x_0, x_1] = \frac{f(x_1) - f(x_0)}{x_1 - x_0}$$

Define $h = f * g$

$$h[x_0, x_1] = \frac{h(x_1) - h(x_0)}{x_1 - x_0} = \frac{f(x_1)g(x_1) - f(x_0)g(x_0)}{x_1 - x_0}$$

Define $\delta = x_1 - x_0, f(x_i) = f_i = f[x_i], g(x_j) = g_j = g[x_j]$:

$$h[x_0, x_1] = \frac{1}{\delta} [f_1 g_1 - f_0 g_0] = \frac{1}{\delta} [f_1 g_1 - f_0 g_1 + f_0 g_1 - f_0 g_0] = \frac{1}{\delta} [g_1 (f_1 - f_0) + f_0 (g_1 - g_0)]$$

$$h[x_0, x_1] = g_1 \frac{f_1 - f_0}{\delta} + f_0 \frac{g_1 - g_0}{\delta} = g[x_1]f[x_0, x_1] + f[x_0]g[x_0, g_0]$$

$$\therefore (fg)[x_0, x_1] = g[x_1]f[x_0, x_1] + f[x_0]g[x_0, g_0]$$

Problem 17

Find the Newton Interpolating Polynomial for these data:

x	1	$\frac{3}{2}$	0	2
$f(x)$	3	$\frac{13}{4}$	3	$\frac{5}{3}$

$$p(x) = \sum_{k=0}^n c_k q_k(x), x_k = F[x_0, x_1, \dots, x_k], q_k(x) = \prod_{j=0}^{k-1} (x - x_j)$$

	$f[]$	$f[,]$	$f[, ,]$	$f[, , ,]$
$x_0 = 1$	3			
		$\frac{1}{2}$		
$x_1 = \frac{3}{2}$	$\frac{13}{4}$		$\frac{1}{3}$	
		$\frac{1}{6}$		-2
$x_2 = 0$	3		$-\frac{5}{3}$	
		$-\frac{2}{3}$		
$x_3 = 2$	$\frac{5}{3}$			

$$\therefore p(x) = 3 + \frac{1}{2}(x-1) + \frac{1}{3}(x-1)\left(x - \frac{3}{2}\right) - 2(x-1)\left(x - \frac{3}{2}\right)(x)$$

$$p(1) = 3$$

$$p\left(\frac{3}{2}\right) = 3 + \frac{1}{2}\left(\frac{3}{2} - 1\right) = 3 + \frac{1}{2}\left(\frac{1}{2}\right) = \frac{13}{4}$$

$$p(0) = 3 + \frac{1}{2}(-1) + \frac{1}{3}(-1)\left(-\frac{3}{2}\right) = 3 - \frac{1}{2} + \frac{1}{2} = 3$$

$$p(2) = 3 + \frac{1}{2}(1) + \frac{1}{3}(1)\left(\frac{1}{2}\right) - 2(1)\left(\frac{1}{2}\right)(2) = 3 + \frac{1}{2} + \frac{1}{6} - 2 = \frac{6}{6} + \frac{3}{6} + \frac{1}{6} = \frac{10}{6} = \frac{5}{3}$$

Problem 19

Show that if u is any function that interpolates f at $x_0, x_1, \dots, x_{n-1}$ and if v is a function that interpolates f at $x_1, x_2, \dots, x_n$, then the function

$$T(x) = \frac{(x_n - x)u(x) + (x - x_0)v(x)}{x_n - x_0}$$

Interpolates f at $x_0, x_1, \dots, x_n$.

We know that $f(x_i) = u(x_i), i = 0, 1, \dots, n$ and $f(x_j) = v(x_j), j = 1, 2, \dots, n$ thus we can deduce from this that $u(x_k) = f(x_k) = v(x_k), k = 1, 2, \dots, n-1$. Thus:

$$\begin{aligned} T(x_k) &= \frac{(x_n - x_k)u(x_k) + (x_k - x_0)v(x_k)}{x_n - x_0} = \frac{(x_n - x_k)f(x_k) + (x_k - x_0)f(x_k)}{x_n - x_0} \\ \Rightarrow T(x_k) &= \frac{x_n f(x_k) - x_k f(x_k) + x_k f(x_k) - x_0 f(x_k)}{x_n - x_0} = \frac{f(x_k)[x_n - x_0]}{x_n - x_0} = f(x_k) \end{aligned}$$

Thus, all we need to check is the endpoints:

$$T(x_0) = \frac{(x_n - x_0)u(x_0) + (x_0 - x_0)v(x_0)}{x_n - x_0} = \frac{(x_n - x_0)u(x_0)}{x_n - x_0} = u(x_0) = f(x_0)$$

$$T(x_n) = \frac{(x_n - x_n)u(x_n) + (x_n - x_0)v(x_n)}{x_n - x_0} = \frac{(x_n - x_0)f(x_n)}{x_n - x_0} = f(x_n)$$

$$\therefore T(x_l) = f(x_l), l = 0, 1, \dots, n$$

Problem 23

The polynomial $p(x) = 2 - (x+1) + x(x+1) - 2x(x+1)(x-1)$ interpolates the first four points in the table:

x	-1	0	1	2	3
y	2	1	2	-7	10

By adding one additional term to p , find a polynomial that interpolates the whole table.

$$p(x) = p_3(x) = 2 - (x+1) + x(x+1) - 2x(x+1)(x-1)$$

$$p_4(x) = p_3(x) + \psi(x), \psi(x) = c_4(x+1)(x)(x-1)(x-2)$$

$$\psi(x_i) = 0, x_i = -1, 0, 1, 2$$

$$p_4(3) = 10 \Rightarrow \psi(3) = 10 - p_3(3)$$

$$\psi(3) = c_4(4)(3)(2)(1) = 24c_4$$

$$p_3(3) = 2 - (4) + (3)(4) - 2(3)(4)(2) = 2 - 4 + 12 - 48 = -38$$

$$\Rightarrow 24c_4 = 10 - (-38) = 48 \Rightarrow c_4 = \frac{48}{24} = 2$$

$$\therefore p_4(x) = 2 - (x+1) + x(x+1) - 2x(x+1)(x-1) + 2(x+1)(x)(x-1)(x+2)$$

$$p_4(3) = -38 + 2(24) = -38 + 48 = 10$$